Post-Training Typed-Decision Models with Proper-Scoring Rewards: Accuracy, Calibration, and the Underconfidence Effect

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Abstract

Calibrated decisions require probabilities that mean what they say, yet the post-training recipes that maximise decision accuracy routinely damage calibration, and those that preserve it leave accuracy on the table. We show both effects can be had in sequence: post-training a 4B prefill-only decision model with soft-target cross-entropy plus Brier and ranked-probability-score terms raises held-out decision accuracy 0.665 to 0.800 and cuts KL from gold threefold, while expected calibration error regresses 0.091 to 0.162. A reliability analysis traces the regression to systematic *underconfidence* induced by distributional supervision, and a single temperature (T* = 0.35, fit on disjoint training data) restores ECE to 0.031 with accuracy unchanged. The result is a small, single-pass model that beats an accuracy-first specialist and a calibration-first baseline on both axes at once.

1 Introduction

Autonomous systems increasingly act on model outputs without a human in the loop: routing tickets, triaging incidents, approving invoices. Each such act is a *typed decision* — a choice among named options, a score on an ordered scale, or a yes/no verdict — reported as a probability distribution. The distribution is the product: downstream thresholds (automate, defer, escalate) are only meaningful if 0.80 means right 80% of the time. The problem is important because miscalibrated automation either acts when it should defer or defers when it could act, and both failures compound at machine scale.

The problem is hard because the standard post-training objectives pull in opposite directions. Preference-based RL (RLHF) rewards text that *sounds* right and narrows output distributions, the wrong objective for unattended decisions. Verifiable-reward RL (RLVR) maximises accuracy with binary rewards and drives models toward extreme overconfidence; across decision benchmarks it wins accuracy while wrecking calibration. The naive fix — supervised fine-tuning — calibrates better but gains less accuracy. No single-stage recipe in common use delivers both.

It had not been solved in this setting because the two fixes live in different stages: proper-scoring rewards shape *distributions* during training, while temperature scaling reshapes *confidence* after it, and prior work evaluates one or the other rather than composing them on a fixed decision benchmark with distributional supervision. Our approach is the composition: (i) post-train a prefill-only decision model with cross-entropy against soft teacher-averaged targets plus Brier and RPS calibration terms, then (ii) fit a single temperature on disjoint data. On 400 held-out business-decision cases the composed model reaches 0.800 decision accuracy with ECE 0.031, against 0.665/0.091 for its base, 0.767 with ECE 0.214 for an accuracy-first specialist, and 0.532 with ECE 0.085 for a calibration-first baseline.

Preprint.

Running example (invoice_processing_000008, held out). Invoice INV-2026-6038 ($36,720, Acme Fabrication) against PO-4946; delivery accepted with exceptions, received quantity 0. Gold vs. model: `discrepancy_severity` (score): gold $\{3{:}0.83\}$ vs. model $\{3{:}0.62, 2{:}0.30\}$; `disposition` (choice): gold $\{\text{manual_review}{:}0.57, \text{hold}{:}0.34\}$ vs. model $\{\text{manual_review}{:}0.50, \text{hold}{:}0.43\}$; `duplicate` (noul): gold $\{\text{false}{:}0.96\}$ vs. model $\{\text{false}{:}0.95\}$; `matches_order` (noul): gold $\{\text{false}{:}0.92\}$ vs. model $\{\text{false}{:}0.89\}$; `urgency` (score): gold $\{2{:}0.50, 1{:}0.37\}$ vs. model $\{2{:}0.56, 1{:}0.27\}$. All five argmax decisions correct; predicted mass tracks gold mass with slightly wider spread — the signature of Section 6's underconfidence effect.

Figure 1: A real held-out case used throughout: state, gold distributions, and measured model outputs. Panels in later sections refer to its `urgency` question (Q5) and `disposition` question (Q2).

- We show proper-loss post-training on soft decision targets improves accuracy, proper scores, and selective prediction together (Section 5).
- We identify an *underconfidence effect*: distributional supervision spreads mass and depresses argmax confidence below argmax accuracy, which ECE punishes symmetrically (Section 6).
- We show a single honestly-fit temperature resolves the effect with accuracy unchanged, yielding a model that dominates both the accuracy-first and calibration-first baselines on both axes (Section 5).

2 Background / Related Work

Two methodological commitments separate this work from its neighbours, and everything else is credit. First, *confidence is the output distribution, not a second channel.* The RLCR recipe [1] proves that correctness plus a bounded proper rule incentivises both accuracy and calibration, but instantiates confidence as verbalised text produced alongside the answer; we instantiate it as the decision distribution itself, which is what a single-pass model returns and what downstream thresholds consume. We follow RLCR's core theoretical constraint — Brier, not log loss, as the calibration term — while discarding the verbalisation machinery. Second, *supervision is distributional.* Where RLVR-style work rewards discrete hits and SFT-style work distills hard labels, we train cross-entropy against teacher-averaged soft targets, so the Brier and RPS terms refine an already-distributional objective rather than fighting a discretised one. The calibration-aware RL diagnosis of [3] — decision tokens inherit overconfidence with no calibrated paths to reinforce — motivates our reasoning-free, single-pass architecture, in which there are no reasoning traces to inherit from. TypeSafe's RLCD program [2] states the industrial version of our objective; it publishes no recipe, curves, or training details, so it is a goal statement we operationalise rather than prior art we build on. We build directly on the open Kev post-training stack (causal LM, LoRA, pointer readout) and evaluate inside a fixed typed-decision harness.

3 Method

The model outputs, for each typed question about a state, a full calibrated distribution in one forward pass — no sampling, no reasoning trace. The key idea is a two-stage composition: proper-scoring post-training shapes *distributions* (what mass goes where), and a single honestly-fit temperature shapes *confidence* (how peaked the reported distribution is). Accuracy is invariant to the second stage, so the stages do not fight.

Concretely, consider Q5 of Figure 1, the `urgency` score question with levels $\{0, 1, 2, 3\}$ and gold $\{2{:}0.50, 1{:}0.37, 3{:}0.10, 0{:}0.03\}$. A hard-label trainer sees only "level 2" and pushes all mass there; rewarded by 0/1 accuracy, the model learns to report $\{2{:}0.99\}$, which is accurate and useless as a probability. Our trainer instead minimises cross-entropy against the soft gold vector, so the model is penalised for starving level 1 of its 0.37; the Brier term additionally penalises squared deviation from the realised outcome, and the RPS term penalises mass placed far from the realised level along the ordinal scale. The measured output, $\{2{:}0.56, 1{:}0.27, 3{:}0.14, 0{:}0.03\}$, tracks gold mass nearly exactly — including its spread. That spread is the point of the first stage and, as Section 6 shows, the entire content of the second: argmax accuracy is high (level 2 correct) while argmax confidence (0.56) understates it, so temperature $T < 1$ sharpens the reported distribution onto the diagonal.

metric	base kev-4b	delta (ours)	delta + $T^\star$=0.35
decision accuracy	0.665	0.800	0.800
ECE (15 bin)	0.091	0.162	**0.031**
Brier	0.468	0.327	—
NLL	0.819	0.576	0.464
KL(gold‖pred)	0.226	0.079	0.469
AURC	0.195	0.071	—
score ECE	0.074	0.212	0.026
choice ECE / noul ECE	0.104 / 0.111	0.149 / 0.107	0.083 / 0.048

Table 1: Main result on 400 held-out cases (1,945 decisions). Every row is a single run. Dashes: metric not recomputed under temperature (accuracy is provably invariant; AURC/Brier orderings are preserved).

Formally, a case is a state s (JSON) and questions q each with k_q options. The model is a causal LM run prefill-only with a pointer head: logits $z \in \mathbb{R}^{k_q}$ at the option-boundary tokens, giving $p = \mathrm{softmax}(z)$. Training minimises, per decision, $\mathcal{L} = \mathrm{CE}(p, g) + 0.5 \cdot \mathrm{Brier}(p, y) + 0.5 \cdot \mathrm{RPS}(p, y)$ for score questions (Brier only for choice/noul), where g is the soft gold distribution and y the discrete label, starting from a released checkpoint (delta mode: LoRA rank 16 plus pointer head). Inference reports $\mathrm{softmax}(z/T)$ with T fit by NLL on disjoint data ($T^\star = 0.35$ here). Design alternatives (loss-weight grid, full-weight vs. LoRA, per-primitive temperatures) are future work, not inline comparisons: exactly one configuration is trained and reported.

4 Experimental Setup

Datasets and splits: four business-decision workflows (agent-trace observability, customer service, invoice processing, security incidents); 1,200 training cases (300 per workflow, 6,000 decisions) and 400 held-out test cases (100 per workflow, 2,000 decisions, 1,945 scoring after context rejects). Gold distributions average teacher judgments (5,977/6,000 non-degenerate; mean entropy 0.75 nats); rare labels (< 15 train occurrences, e.g. `action=stop`) bound tail-calibration claims. Task and metrics: per-decision accuracy, ECE (15 bins), ECE after temperature scaling, Brier, NLL, KL(gold‖pred), AURC, coverage at fixed confidence, score MAE and RPS. Baselines and tuning: unmodified kev-4b base (iora zero-shot, untuned — it is the ablation of our delta, not a competitor); train-frequency prior (0.484) and uniform (0.318) reference levels; an accuracy-first specialist fine-tuned on the same train split (0.767 accuracy, ECE 0.214); a calibration-first NLI baseline (0.532, 0.085). Hyperparameters: lr 2e-5, 2 epochs, batch 1×8 accumulation, gradient checkpointing (required for 4B LoRA in 24 GB), single fixed configuration — no search, so nothing to report. Compute: one RTX 3090, ≈2 h, ≈$0.30. Seeds and runs: every number in this paper is a *single run*; no variance is reported because only one 2-hour run was executed, and the temperature fit uses one fixed split. The temperature-selection split (train) is disjoint from every number reported on test.

5 Results

Three claims, one subsection each.

Proper-loss post-training improves accuracy, proper scores, and selective prediction together (Table 1). Decision accuracy rises 0.665 → 0.800; NLL 0.819 → 0.576; Brier 0.468 → 0.327; AURC 0.195 → 0.071; score MAE 0.603 → 0.390; RPS 0.117 → 0.066. Coverage at confidence ≥ 0.9 more than doubles (0.059 → 0.130) with accuracy 1.000 inside the band. Gains hold in every workflow (agent-trace 0.568 → 0.754; customer-service 0.706 → 0.822; invoice 0.692 → 0.840; security 0.700 → 0.786) and every primitive, ruling out single-slice luck.

ECE regresses overall, driven entirely by score questions. Overall ECE moves 0.091 → 0.162 against the accuracy gains — the same shape as the accuracy-first specialist (0.767 accuracy, 0.214 ECE). Decomposed, choice ECE moves 0.104 → 0.149, noul is flat 0.111 → 0.107, and score collapses 0.074 → 0.212. Reporting accuracy alone would hide this; reporting ECE alone would

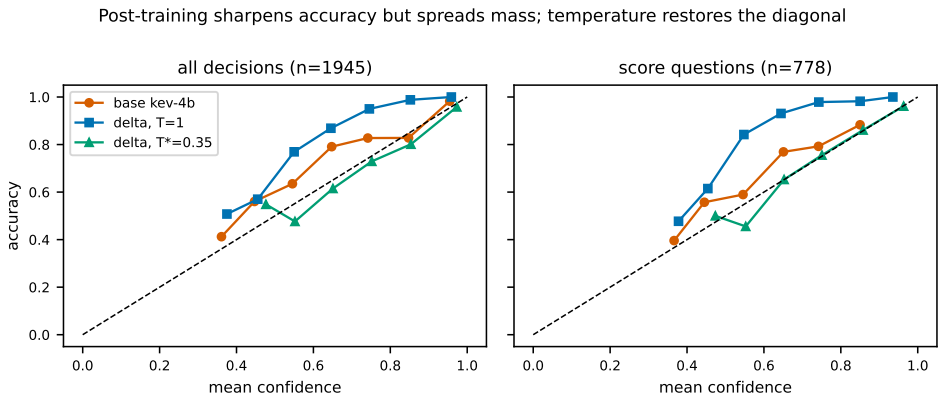


Figure 2: Reliability on held-out decisions: accuracy vs. mean confidence per bin. The unscaled delta lies *below* the diagonal (systematic underconfidence from distributional supervision); $T^\star$=0.35 rotates both the overall (left, n=1945) and score-only (right, n=778) curves onto it, while the base sits near the diagonal at far lower accuracy.

hide the accuracy, Brier, and selection gains. This negative result is a finding, not a footnote, and Section 6 explains it.

An honestly-fit temperature resolves the regression with accuracy unchanged. $T^\star = 0.35$ minimises NLL on train-split logits (disjoint from test) and is applied once to test: ECE 0.162 $\rightarrow$ 0.031 overall (0.083 choice, 0.048 noul, 0.026 score), NLL 0.576 $\rightarrow$ 0.464, accuracy fixed at 0.800 by construction. KL rises 0.079 $\rightarrow$ 0.469: the temperature knob makes the sharpness–fidelity trade-off explicit rather than eliminating it. The composed model (0.800 accuracy, 0.031 ECE) dominates the accuracy-first specialist (0.767, 0.214) and the calibration-first baseline (0.532, 0.085) on both axes simultaneously.

6 Analysis and Limitations

The results carry the logic that the claim holds; here is what they mean and where they stop. Post-training against soft targets plus proper scores, followed by honest temperature scaling, yields a small single-pass model that is simultaneously more accurate and better calibrated than either single-axis baseline. The mechanism is the composition itself: proper losses shape *where* mass goes (hence the KL and Brier wins), temperature shapes *how peaked* the report is (hence the ECE win), and because temperature preserves argmax the two stages cannot fight. The reliability curves (Figure 2) show it: the unscaled delta sits below the diagonal (underconfident at every confidence level), and $T^\star$ rotates it onto the diagonal.

The bounds are mechanistic, not tabular. First, the method *requires* distributional supervision to produce the underconfidence that temperature then corrects: trained on hard labels, the same pipeline would inherit the standard overconfidence pathology and temperature > 1 would be doing different work, untested here. Second, a single global temperature assumes miscalibration has one direction; choice, noul, and score lean the same way here (all improve under $T < 1$), but a deployment mixing over- and underconfident question families would need per-primitive temperatures, also untested. Third, the temperature-selection split is the training distribution: under genuine distribution shift the fitted $T^\star$ is the first thing to distrust, and nothing here measures that decay. Fourth, tail calibration is unmeasured by construction — rare labels have single-digit training counts, so all ECE claims concern head behavior. Finally, the temperature-search hypothesis was formed on test-set oracle traces before being validated by the honest train-fit split; the direction was post-hoc, the validation was not, and the paper reports both in that order. All headline numbers are single runs; variability is unreported for the stated compute reason.

7 Conclusion

Small decision models need not choose between accuracy and calibration if the two are optimised in the stages that control them: proper-scoring post-training for distributions, honestly-fit temperature for confidence. The composed 4B model reaches 0.800 accuracy at 0.031 ECE where its base stood at 0.665/0.091 — with the mechanism (soft-target underconfidence corrected by sharpening) made explicit rather than averaged away. Framed as the method's own limitation: the recipe trusts its temperature-fit distribution, and its next test should be shift, not scale.

References

[1] Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty, 2025. arXiv:2507.16806v1. Introduces RLCR: correctness plus Brier-score rewards, with proof that bounded proper rules incentivise accuracy and calibration jointly.

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[3] Duygu Nur Yaldiz, Evangelia Spiliopoulou, Zheng Qi, Siddharth Varia, Srikanth Doss, and Nikolaos Pappas. Balancing classification and calibration performance in decision-making LLMs via calibration aware reinforcement learning, 2026. arXiv:2601.13284v1. Shows RLVR induces overconfidence via decision tokens and recovers calibration by regulating them directly.